

Servo Motor Tester

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When using a servo motor in a project, if the servo motor does not respond as per the input, how to make sure that the fault is not in the servo motor but the circuit or logic? One way is to isolate the servo motor from the circuit and check its proper working by feeding it

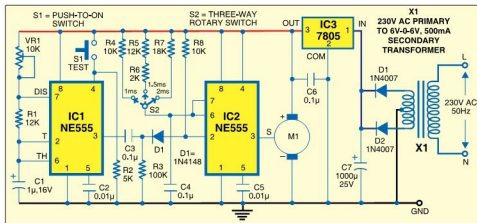
pulses of varying width and checking the angle that the servo motor turns to. For example, a 1.5ms pulse should make the motor turn to a 90-degree position (neutral position).

The circuit presented here generates pulses of varying widths. It is built around two NE555 timer ICs (IC1 and IC2) and a few discrete components. Timer IC1 is configured as an astable

multivibrator with a time period of 20 ms. Every 20 ms, the astable provides a very sharp negative pulse to trigger IC2. Timer IC2 is configured as a monostable multivibrator that produces 1ms, 1.5ms and 2ms long pulses to rotate the servo motor (M1).

Pin 4 of IC1 is pulled down by resistor R2. When switch S1 is pressed, the astable multivibrator triggers the monostable to produce a pulse as per the position of switch S2. Switch S2 can select resistors R4, R5 and R6 together, and R7 to produce monostable pulse output of 1 ms, 1.5 ms and 2 ms, respectively. Preset VR1 is used to set the time period of IC1 to 20 ms.

Using switch S2, select the monostable time period as 1 ms, 1.5 ms or 2 ms and press switch S1. The servo motor should rotate to extreme left, middle or extreme right, respectively. ●



1Hz Precision Clock Timer

RAJ K. GORKHALI

This timer makes an LED flash once every second along with a beep. The timing circuit uses a quartz crystal available in digital watches or clocks which generates precise frequency of 32.768 kHz.

The circuit comprises a 14-stage ripple-carry binary counter and oscillator IC CD4060 (IC1), dual flip-flop CD4013 (IC2), transistors T1, T2 and T3, a speaker, a quartz crystal and a few discrete components. The inbuilt inverter of IC1 (which is between pins 10 and 11) in combination with resistors, capacitors and the quartz crystal forms the oscillator circuit.

The clock signal of 32.768 kHz

has a pulse width of about 30 microseconds, which is far too short to be able to count the oscillations. So this frequency is fed to IC1, which divides

it by 14 to output a frequency of 2 Hz.

The 2Hz output is fed to IC2, which divides it by 2, giving a final output of

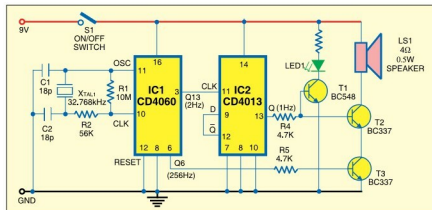


Fig. 1: 1Hz precision clock timer